

ANX-PR/CL/001-01

LEARNING GUIDE

SUBJECT

103001022 - Software Engineering For Deep Learning

DEGREE PROGRAMME

10AM - Master Universitario En Ingenieria Del Software

ACADEMIC YEAR & SEMESTER

2025/26 - Semester 1

Index

Learning guide

1. Description.....	1
2. Faculty.....	1
3. Prior knowledge recommended to take the subject.....	2
4. Skills and learning outcomes	2
5. Brief description of the subject and syllabus.....	4
6. Schedule.....	5
7. Activities and assessment criteria.....	7
8. Teaching resources.....	10
9. Other information.....	10

1. Description

1.1. Subject details

Name of the subject	103001022 - Software Engineering For Deep Learning
No of credits	4 ECTS
Type	Optional/elective
Academic year of the programme	First year
Semester of tuition	Semester 1
Tuition period	September-January
Tuition languages	English
Degree programme	10AM - Master Universitario en Ingeniería del Software
Centre	10 - E.T.S. De Ingenieros Informáticos
Academic year	2025-26

2. Faculty

2.1. Faculty members with subject teaching role

Name and surname	Office/Room	Email	Tutoring hours *
Maria Pilar Rodriguez Gonzalez	6203	pilar.rodriguez@upm.es	W - 14:00 - 17:00 F - 12:00 - 15:00
Sira Vegas Hernandez (Subject coordinator)	5105	sira.vegas@upm.es	M - 12:00 - 15:00 Th - 12:00 - 15:00

* The tutoring schedule is indicative and subject to possible changes. Please check tutoring times with the faculty member in charge.

3. Prior knowledge recommended to take the subject

3.1. Recommended (passed) subjects

The subject - recommended (passed), are not defined.

3.2. Other recommended learning outcomes

- Basic knowledge of machine learning (recommended)
- Basic knowledge of software engineering (essential)
- Python programming language (essential)

4. Skills and learning outcomes *

4.1. Skills to be learned

CE13 - Tener una visión de los distintos aspectos específicos y emergentes de la ingeniería del software, y profundizar en algunos de ellos

CE14 - Comprender lo que pueden y no pueden conseguir las prácticas actuales de ingeniería del software, y sus limitaciones y su posible futura evolución.

CG1 - Que los estudiantes sepan aplicar los conocimientos adquiridos y su capacidad de resolución de problemas en entornos nuevos o poco conocidos dentro de contextos más amplios (o multidisciplinares) relacionados con su área de estudio (RD)

CG13 - Apreciación de los límites del conocimiento actual y de la aplicación práctica de la tecnología más reciente

CG14 - Conocimiento y comprensión de la informática necesaria para la creación de modelos de información, y de los sistemas y procesos complejos

CG18 - Capacidad de trabajar y comunicarse también en contextos internacionales

4.2. Learning outcomes

RA1 - Within an application field of Software Engineering, uses and designs the appropriate solution to solve some of its problems, describing the technical difficulties and the application limits

RA23 - Time organization capability SC13, SC14 K

RA25 - Communication skills in public SC13, SC14, CG3, CG18 S

RA84 - Being able to reframe a business question as a data question, reasoning about what data might be of assistance and how to obtain it

RA54 - Proponer una solución justificada a un problema real que sea complejo o mal definido, o perteneciente a un área nueva o emergente, o que requiera el desarrollo de enfoques o métodos nuevos y originales, dentro del contexto de la ingeniería del software justificándola de una forma cualitativa y cuantitativa.

RA56 - Materializar la solución propuesta a un problema dado en términos de código, prototipo, informes, pruebas de concepto, análisis, diseños y/o documentación, ubicándola en un entorno empresarial real.

RA18 - Given a real problem, the student chooses the most appropriate software engineering solution, analyzing the solution feasibility, what can and cannot be achieved through the current status of the chosen solution, and what it can advance in the future.

RA19 - The student explains what are the software engineering limits and frontiers, and the base for new trends and developments, and about the advanced issues and their application.

RA26 - Group work skill SC13, SC14, CG17 A

* The Learning Guides should reflect the Skills and Learning Outcomes in the same way as indicated in the Degree Verification Memory. For this reason, they have not been translated into English and appear in Spanish.

5. Brief description of the subject and syllabus

5.1. Brief description of the subject

Deep learning (DL) is emerging as a powerful approach to address very complex problems, including those faced by software engineers. It is now being explored, for example, to automatically document code, automate test generation, and even provide code patches. Yet, the DNNs being developed and their validation are at their infancy, suffering from many limitations that we do not quite understand, which may impact future research and their adoption in practice.

This course aims at understanding how DL can be incorporated to the development of software systems, and in which types of tasks it can be used.

5.2. Syllabus

1. Introduction to Deep Learning
 - 1.1. What is DL?
 - 1.2. Basics of DL
 - 1.3. Training Deep Feedforward Networks
 - 1.4. Neural Networks Architectures
2. Deep Learning for Software Engineering
 - 2.1. SE Problems that Use DL
 - 2.2. DL Architectures Used to Solve SE Problems
 - 2.3. Integrating DL into SE
3. Software Engineering for Deep Learning
 - 3.1. Developing a DL System
 - 3.2. SE Challenges of DL
 - 3.3. Incorporating SE practices to DL

6. Schedule

6.1. Subject schedule*

Week	Type 1 activities	Type 2 activities	Distant / On-line	Assessment activities
1	Chapter 1: Description of the course and assignment Duration: 02:00 Lecture			
2	Chapter 1: Introduction to DL Duration: 02:00 Lecture			
3	Chapter 1: Introduction to DL Duration: 02:00 Laboratory assignments			
4	Chapter 1: Introduction to DL Duration: 02:00 Laboratory assignments			
5	Chapter 2: SE for DL Duration: 02:00 Lecture			
6	Chapter 2: SE for DL Duration: 02:00 Lecture			
7	Presentation of Assignment: Milestone 1 Duration: 02:00 Additional activities			Draft Submission of Assignment: Milestone 1 Group work Progressive assessment Not Presential Duration: 10:00 Presentation of Assignment: Milestone 1 Group presentation Progressive assessment Presential Duration: 02:00
8	Chapter 2: SE for DL Duration: 02:00 Lecture			Submission of Assignment: Milestone 1 Group work Progressive assessment Not Presential Duration: 01:00
9	Follow up of Assignment: Milestone 2 Duration: 02:00 Cooperative activities			
10	Chapter 2: SE for DL Duration: 02:00 Lecture			

11	Presentation of Assignment: Milestone 2 Duration: 02:00 Additional activities			Draft Submission of Assignment: Milestone 2 Group work Progressive assessment Not Presential Duration: 20:00 Presentation of Assignment: Milestone 2 Group presentation Progressive assessment Presential Duration: 02:00
12	Follow up of Assignment: Milestone 3 Duration: 02:00 Cooperative activities			Submission of Assignment: Milestone 2 Group work Progressive assessment Not Presential Duration: 01:00
13	Chapter 3: DL for SE Duration: 02:00 Lecture			
14	Presentation of Assignment: Milestone 3 Duration: 02:00 Additional activities			Presentation of Assignment: Milestone 3 Group presentation Progressive assessment Presential Duration: 02:00 Draft Submission of Assignment: Milestone 3 Group work Progressive assessment Not Presential Duration: 30:00
15	Submission of Assignment: Milestone 3 Duration: 02:00 Additional activities			Submission of Assignment: Milestone 3 Group work Progressive assessment Presential Duration: 01:00
16				Attendance Individual work Progressive assessment Presential Duration: 00:00 Re-submission of Assignment (all 3 milestones) Group work Global examination Presential Duration: 60:00
17				

Depending on the programme study plan, total values will be calculated according to the ECTS credit unit as 26/27 hours of student face-to-face contact and independent study time.

7. Activities and assessment criteria

7.1. Assessment activities

7.1.1. Assessment

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
7	Draft Submission of Assignment: Milestone 1	Group work	No Presential	10:00	20%	3 / 10	CG13 CG14 CE14 CG1 CE13
7	Presentation of Assignment: Milestone 1	Group presentation	Face-to-face	02:00	5%	0 / 10	CG18
8	Submission of Assignment: Milestone 1	Group work	No Presential	01:00	20%	3 / 10	CG13 CG14 CE14 CG1 CE13
11	Draft Submission of Assignment: Milestone 2	Group work	No Presential	20:00	30%	3 / 10	CG14 CE14 CG13 CG1 CE13
11	Presentation of Assignment: Milestone 2	Group presentation	Face-to-face	02:00	5%	0 / 10	CG18
12	Submission of Assignment: Milestone 2	Group work	No Presential	01:00	30%	3 / 10	CG13 CG14 CE13 CG1 CE14
14	Presentation of Assignment: Milestone 3	Group presentation	Face-to-face	02:00	5%	0 / 10	CG18
14	Draft Submission of Assignment: Milestone 3	Group work	No Presential	30:00	30%	3 / 10	CG13 CG14 CE14 CG1 CE13
15	Submission of Assignment: Milestone 3	Group work	Face-to-face	01:00	30%	3 / 10	CG13 CG14 CE14 CG1 CE13

16	Attendance	Individual work	Face-to-face	00:00	5%	3 / 10	CG18
----	------------	-----------------	--------------	-------	----	--------	------

7.1.2. Global examination

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
16	Re-submission of Assignment (all 3 milestones)	Group work	Face-to-face	60:00	80%	5 / 10	CE13 CG1 CG13 CG14 CE14

7.1.3. Referred (re-sit) examination

Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
Second re-submission of Assignment (all milestones)	Group work	Face-to-face	60:00	80%	5 / 10	CG1 CG13 CG14 CE13 CE14

7.2. Assessment criteria

Progressive evaluation period:

- No examination will be made.
- Students will be evaluated using the evaluation tasks described above. Each of the 3 milestones of the assignment has two submissions associated: a draft submission (compulsory) and a consolidated submission (optional). The draft submissions will be presented in class in voluntary basis. Based on the feedback received by the students from the professor the day of the presentation, the students will have the opportunity to fix the draft version and submit a consolidated version.
- The final grade will be calculated using a weighted average as follows:
 - If there is a consolidated version of the milestone, the consolidated version will be graded.

Otherwise, the draft version will be graded.

- The weights will be computed according to the information given above.
- Presentations and attendance are non-recoverable activities, due to their nature. They formers are presentations of the submitted assignment and the latter computes the attendance to the course.

Global evaluation:

When the overall score obtained by the student in the progressive evaluation period is smaller than 5, the student will have to re-submit those milestones of the assignment that do not reach the minimum score required. It is not possible to complete milestones for which there is not a submission in the progressive evaluation period. In no case milestones that have a score equal or greater than 5 will be repeated. The score for the presentations and attendance will be taken from the progressive evaluation period (as they are non-recoverable activities).

Extraordinary evaluation:

When the overall score obtained by the student in the global evaluation period is smaller than 5, the student will have to repeat (or perform in case (s)he has not done it before) the milestones that do not reach the minimum score required. In no case milestones that have a score equal or greater than 5 will be repeated. The score for the presentations and attendance will be taken from the progressive evaluation period (as they are non-recoverable activities).

8. Teaching resources

8.1. Teaching resources for the subject

Name	Type	Notes
Deep Learning. I. Goodfellow, Y. Bengio, A. Courville. The MIT Press	Bibliography	
Neural Networks and Deep Learning. C.C. Aggarwal. Springer	Bibliography	

9. Other information

9.1. Other information about the subject

En la asignatura se implementa la siguiente metodología docente innovadora (<https://innovacioneducativa.upm.es/guias-pdi>) con el fin de motivar y reforzar el aprendizaje por parte del estudiantado:

- **Aprendizaje orientado a proyectos:** La evaluación de la asignatura se realiza a través de un proyecto, que los alumnos realizarán en pequeños grupos, relativo al desarrollo de un sistema basado en aprendizaje profundo siguiendo prácticas de Ingeniería del Software.